Check for updates

Received: 3 May 2022 | Revised: 19 July 2022 | Accepted: 13 August 2022

DOI: 10.1002/for.2909

RESEARCH ARTICLE

WILEY

Text-based soybean futures price forecasting: A two-stage deep learning approach

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Funding information
National Social Science Foundation of China, Grant/Award Number: 20&ZD126

Abstract

This paper investigates the soybean futures price prediction problem from a new perspective and proposes an effective prediction model named Two-Stage Hybrid Long Short-Term Memory (TSH-LSTM) by using text data from social media. First, the unstructured text is transformed into structured data by sentiment analysis and text classification methods. The improved sentiment score is computed by combining the degree centrality of sentiment words based on the sentiment dictionary method, and the characteristics of price fluctuations in texts are learned through the text Recurrent Convolutional Neural Networks. Second, the significant relationship between social media features and soybean futures price is assessed through stepwise regression, and the results of such an assessment are used as a basis for the identification of significant factors as input variables of the prediction model. Finally, the TSH-LSTM prediction model is designed, and the final prediction result is acquired through the combination of prediction results of each stage using the error reciprocal method. The empirical results indicate that the incorporation of the social media text feature helps improve forecasting performances. Specifically, the proposed TSH-LSTM is more accurate than univariate LSTM, multivariate LSTM, and eXtreme Gradient Boosting.

KEYWORDS

soybean futures price forecast, stepwise regression, text, two-stage LSTM

1 | INTRODUCTION

Soybean futures price forecast plays an increasingly vital role in food policy making and trade risk management (Bai & Xue, 2021). However, the influence factors of soybean futures price are complex and diverse, which makes the price fluctuate violently and poses a great challenge to the prediction of the soybean futures price.

Statistical methods are mainly employed in the early stage of soybean futures price prediction due to the restrictions of data acquisition and analysis methods to examine the features of historical price series and create prediction models (Papaias & Thomakos, 2017). With the rapid development of artificial intelligence technology, the prediction accuracy of deep learning models is significantly better than that of statistical models (Lv & Wang, 2022; Peng et al., 2022). Among them, LSTM and its improved version are widely used in agricultural product price prediction. For example, Gu et al. (2022) propose dual input attention long short-term memory (DIA-LSTM) for the efficient prediction of agricultural commodity prices. Jaiswal et al. (2022) developed a deep long short-term memory (DLSTM) based model for the accurate forecasting of agricultural price series. The experimental results in these studies all demonstrate that LSTM-based models exhibit better performance than

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existing methods, in terms of various forecasting evaluation criteria like root mean square error, mean absolute percentage error, and mean absolute deviation. However, these models are based on time-series data, ignoring the significance of external influencing factors, such as extreme weather, emergencies, and sentiment in agricultural markets. Moreover, the measurement of these factors is still an open question (Adämmer & Bohl, 2018; Ahumada & Cornejo, 2016a; Bakas & Triantafyllou, 2019).

Fortunately, Novel opportunities for soybean futures price prediction are introduced by the rapid development of artificial intelligence and big data technology. The information on social media such as futures topics on Weibo and information on soybean futures trading websites, comprises a considerable amount of untapped qualitative information and can be used to improve the prediction of soybean futures prices (Brynjolfsson et al., 2016; Ceron et al., 2014; Eslami et al., 2018). For example, Chuluunsaikhan et al. (2020) combined LSTM and topic modeling techniques to predict the daily retail price of pork in the Korean domestic market based on news articles. Li et al. (2022) extract the topic distribution and perform sentiment analysis on news headlines, using sentiment scores as proxy variables for difficult-to-measure variables such as policy and weather. The above studies show that variables constructed from the text can improve the accuracy of predictive models. Despite some progress obtained in current agricultural product price forecasting studies, there are still some concerns that need to be addressed in future studies. (a) The sentiment analysis method in the current research does not consider the influence of domain sentiment vocabulary, and the accuracy of sentiment classification needs to be improved; (b) Topic modeling techniques identify the surface features in the text, and the deep features in the text need to be further excavated using deep learning methods.

Therefore, the motivation of the paper is to build an LSTM-based two-stage prediction model combined with textual data. Figure 1 presents the flowchart of this study. First, we combined social network analysis and a sentiment dictionary to perform sentiment analysis on Weibo text. Taking price fluctuations as labels, we use textRCNN (text Recurrent Convolutional Neural Networks) to learn the deep features of price fluctuations in the texts. Second, combined with statistical variables, use stepwise regression analysis to perform feature selection, and obtain the optimized feature set. Finally, a two-stage hybrid prediction model named TSH-LSTM is proposed. The univariate LSTM, multivariate LSTM, and XGBoost are used as comparison models to verify the effectiveness of the proposed prediction model. The contributions are as follows.

a. The proposed TSH-LSTM based on econometric analysis and deep learning for the soybean futures price forecasting can take full advantage of the univariate

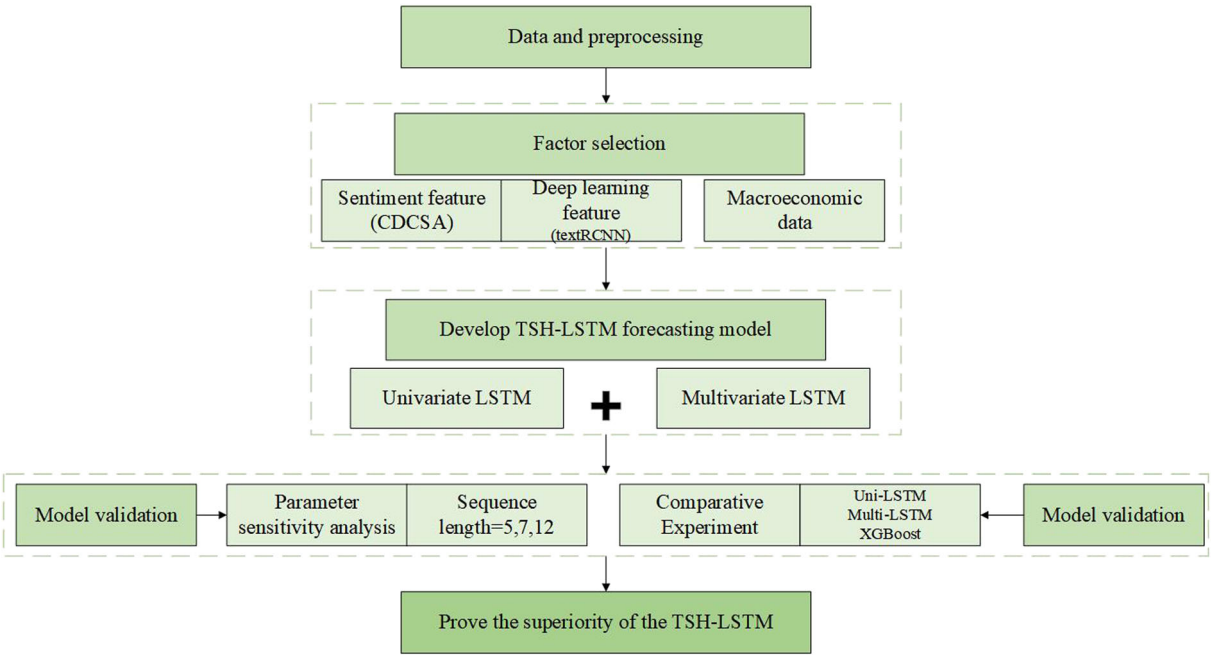

FIGURE 1 Flowchart of soybean futures price forecasting

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and multivariable time series forecasting methods, the performance of the proposed TSH-LSTM is further enhanced by using the error reciprocal method. The mean absolute percentage error for soybean futures price forecast is below 5.9%.

b. Soybean price prediction accuracy is further improved through the mutual supplement of deep features constructed by textRCNN and emotional features constructed by the proposed CDCSA (Combining Degree Centrality and Sentiment Analysis) method. Specifically, the proposed CDCSA method by combining social network analysis and sentiment dictionary improves the accuracy of sentiment analysis and constructs sentiment variables more efficiently.

The remainder of this paper is arranged as follows. Related research on Weibo sentiment analysis and soybean futures price prediction is presented in Section 2. The research framework and algorithm principles are highlighted in Section 3. The process of text feature extraction, factor selection, and soybean futures price forecast evaluation is introduced in Section 4. The theoretical and practical contributions, limitations, and future research directions are highlighted in Section 5.

2 | LITERATURE REVIEW

2.1 | Text mining method in agricultural price prediction

With the rapid development of natural language processing technology, text data has been recognized as an important indicator for agricultural futures price prediction (Devyatkin & Otmakhova, 2021). A series of studies focus on constructing textual features using text mining methods and then adopting the combination of textual and non-textual factors for forecasting (Ling et al., 2020).

Twitter data are widely used in the research on price predictions of agricultural commodities (Kim et al., 2017; Pulse, 2014; Surjandari et al., 2015) because millions of users share their opinions on this platform. Sentiment analysis is one of the most commonly used text feature mining methods. For example, Surjandari et al. (2015) used three machine learning algorithms (i.e., Naïve Bayes, Support Vector Machine [SVM], and Decision Tree) to perform sentiment analysis on tweets data, and then used chi-square tests and the Marascuillo procedure to analyze associations between staple types and sentiment categories. The results show that the price of milk, eggs, and red onions are most significantly associated with negative sentiment compared to other commodities, while support vector machine (SVM) classifiers yield higher accuracy than naive Bayes and decision trees. Kim et al. (2017) proposed a two-step algorithm for the "nowcasting" of commodity prices using social media. They extract tweets mentioning price quotations of four food commodities using sentiment analysis algorithms, then build a statistical model to nowcast the prices. The result shows that when the number of tweets about food prices increase, food prices change sharply. Topic modeling is another commonly used text analysis method. For example, Chuluunsaikhan et al. (2020) combined LSTM and topic modeling techniques to predict the daily retail price of pork in the Korean domestic market based on news articles. They used a topic modeling technique to obtain relevant keywords that can express price fluctuations. Based on these keywords, they constructed prediction models using statistical, machine learning, and deep learning methods. The experimental results show that there is a strong relationship between the meaning of news articles. Li et al. (2022) extract the topic distribution of news headlines to obtain a series of comprehensive influencing factors, and then perform sentiment analysis on news headlines, using sentiment scores as proxy variables for difficult-to-measure variables such as policy and weather. The experimental results show that the variables identified from the text can significantly improve the performance of predictive models.

The above research shows that text data is of great significance to agricultural product price prediction, and can significantly improve the prediction effect of the model. Nonetheless, the text feature mining method in the text-based agricultural futures price prediction research still has some limitations. The accuracy of sentiment analysis needs to be further improved due to a large amount of redundant information in the text and the domain-independent sentiment vocabulary. Sentiment analysis methods and topic modeling methods obtain the surface features of the text, and the deep features in the text need to be mined by advanced deep learning methods. Through the above analysis, the paper combines social network analysis and sentiment dictionary to perform sentiment analysis on Weibo texts, to improve the accuracy of sentiment classification; meanwhile, textRCNN is used to extract the deep features of texts.

2.2 | Forecast of agricultural commodity price

The empirical research on soybean futures price prediction comprises two main directions. The first direction emphasizes the fluctuation feature analysis of agricultural futures prices based on the time series forecasting method, disregarding other exogenous forecasting factors

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(Fang et al., 2018). For example, VECM and MSVR were utilized by Xiong et al. (2015) to obtain the linear and nonlinear features of agricultural futures prices and achieve the interval prediction of these prices. The original sequence was decomposed by Wang and Li (2018) into independent components of different scales through singular spectrum analysis, and an SSA causality test was presented to examine the relationships of commodity futures prices. Their results revealed the superiority of the neural network model with SSA to the benchmark model in various measures. Finding a model that realizes the optimal prediction performance in all cases is challenging despite the good prediction performance of the artificial intelligence model for highly complex data (Sezer et al., 2020). Therefore, the application of the hybrid model method is suggested. For example, the volatility model and the volatility prediction of energy futures contracts were respectively achieved by Alizadeh et al. (2019) through the combination of heterogeneous autoregression and the Markov system transformation method. Their results demonstrate the superiority of the MRS-HAR-RV model in obtaining the dynamic of energy futures price volatility to the alternative model. The combination of variational mode decomposition (VMD) and artificial neural network (ANN) and the conversion of high volatility futures price prediction problem into a multicomponent time series prediction problem with a unique center frequency were realized by Liu et al. (2021). They found substantially better numerical and trend accuracies of VMD-ANN than some of the latest methods for energy and metals futures. Motivated by economic theory, the second direction discusses the crucial participation of certain fundamental or price drivers in clarifying changes in agricultural futures prices. For example, the volatility connectivity between the US crude oil futures and Chinese agricultural futures and the existence of volatility spillover between the US crude oil market and Chinese agricultural futures were respectively investigated and validated by Luo and Ji (2018). A multifactor model embedded with seasonal components was developed by Schneider and Tavin (2021) to aid in the volatility prediction of futures prices of agricultural products in the market. They found that the seasonal model is superior to the non-seasonal model in five markets, thus presenting predicting highly accurate predictions of volatility peaks of four commodities. A hybrid classification framework for price trend prediction, which incorporates cross-field open data to predict the price trend of bulk commodities in the next few days, including supply factors of commodity supply sources, demand factors of shipping transport, and air quality, was presented by Zhou et al. (2020). The prediction accuracy of the framework is superior to several state-of-the-art baselines based on their results. A multisource heterogeneous data analysis method for palm oil futures price prediction was introduced by Chai et al. (2020). This method can measure news event data and investor comments using an established relationship graph and sentiment dictionary and model the quantified multisource heterogeneous information with an extended hidden Markov model.

These studies still have limitations despite their advancements in soybean futures price prediction. Effective prediction of future changes in variables can be realized through the time series prediction model, which can learn the features, trends, and development rules of the research object in a certain period. The prediction curve can effectively fit the price curve trend in the short term under general conditions but performs poorly in numerous details, and the long-term prediction effect is not ideal. The traditional time series model excessively concentrates on time series and disregards the effect of external conditions. Thus, this model results in large deviations with remarkable variations of external factors. Such deviations endow the model with large prediction defects, especially for soybean, which has different characteristics, including food, feed, and energy crops. The multivariable time series prediction model not only fits the trend of the entire price curve but also some short-term fluctuations of the curve under multiple factors. Consequently, the inclusion of additional features to simple time series contributes to the continuous improvement of prediction by numerous researchers. A high-quality feature set is an important prerequisite for the ideal effect of multivariate prediction models, so it is necessary to further optimize the feature construction method.

Therefore, this paper proposes a two-stage LSTM prediction model combined with a text mining method. In the feature construction stage, textRCNN and the proposed CDCSA method are used to construct sentiment features and deep features from microblog texts. Combined with statistical variables, the stepwise regression analysis method is used for feature selection. In the prediction stage, the prediction results of the univariate LSTM and the multivariate LSTM are added as the final prediction result by the reciprocal error method.

3 | RESEARCH FRAMEWORK

3.1 | Theoretical foundation

Behavioral finance theory studies the influence of behavioral and psychological factors on relevant decision-making areas, indicating that public sentiment affects price fluctuations in financial markets to a certain extent

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(Colladon et al., 2019; Dolan, 2002; Feuerriegel & Gordon, 2018). The theory of behavioral finance can provide a novel analytical perspective for predicting soybean futures prices. Meanwhile, information theory highlights the dependence of the transformation process of data into information on certain technical acquisition capabilities, that is, data processing and analysis. Extracting valuable features from unstructured data of social media through deep learning is a promising way to improve the performance of prediction models (Wu Gandomi & Haider, 2015). Therefore, a novel research framework for the prediction of soybean futures prices is introduced in this paper through the combination of behavioral finance and information theories.

3.2 | System design of soybean futures price forecast

The framework of the soybean futures price prediction model introduced in this paper, which is divided into four stages, is presented in Figure 2.

Data collection and preprocessing are included in the first stage. The collection of data related to soybean futures prices from multiple sources, which include daily trading, macroeconomic, and social media text data, as well as data cleaning, feature coding, dimensionality reduction, normalization, and other processing methods, are performed.

Data extraction of soybean futures text on social media is the second stage. First, the daily sentiment score is computed by using a new method of sentiment analysis combined with degree centrality (CDCSA), and the daily sentiment score is used as a basis for the sentiment curve within the sample range. Second, the features of price rise and fall in soybean futures text are learned and classification is realized through the chosen advanced deep learning algorithms, which include textCNN, textRCNN, and other models.

The third stage uses a stepwise regression method for feature selection. Independent variables are social media features (including sentiment features and depth features) and macroeconomic features, and the dependent variables are soybean futures prices. The features that have a significant impact on soybean futures prices are taken as optimal features.

The fourth stage builds a two-stage hybrid prediction model based on LSTM. First, a univariate LSTM prediction model based on historical price series is established. Second, a multivariate LSTM prediction model is established. The reciprocal error method finally combines two-stage forecasts for the final forecast.

3.3 | Social media text feature mining

3.3.1 | Sentiment features constructed by CDCSA

1. Social network analysis and Degree Centrality

Social network analysis is a research method for quantitative analysis of social networks. It reveals the characteristics of individuals, groups, and even the entire network through the quantitative representation of various relationships in the network, and analyzes and evaluates node influence and diffusion process. Degree Centrality (DC) normalizes the degrees of nodes according to the total number of nodes in the network.

Assuming that there are N nodes in the network G, among which, the node x has a total of k nodes directly connected to it, then the degree of the node is denoted as $deg(x) = k$, and the corresponding degree centrality is calculated as Equation (1):

$$D_G(x) = \frac{deg(x)}{N-1} \tag{1}$$

where nodes represent sentiment words in the text, and the higher the degree of centrality of sentiment words, the stronger the sentiment polarity. Incorporating social network information from Weibo texts into sentiment

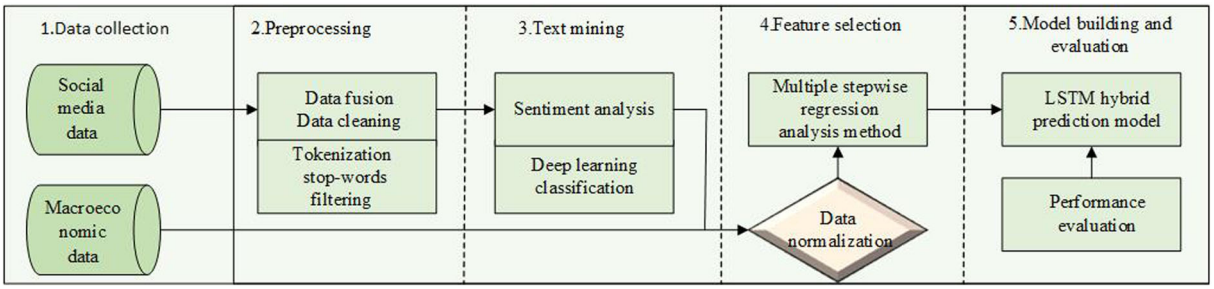

FIGURE 2 System design of soybean futures price prediction

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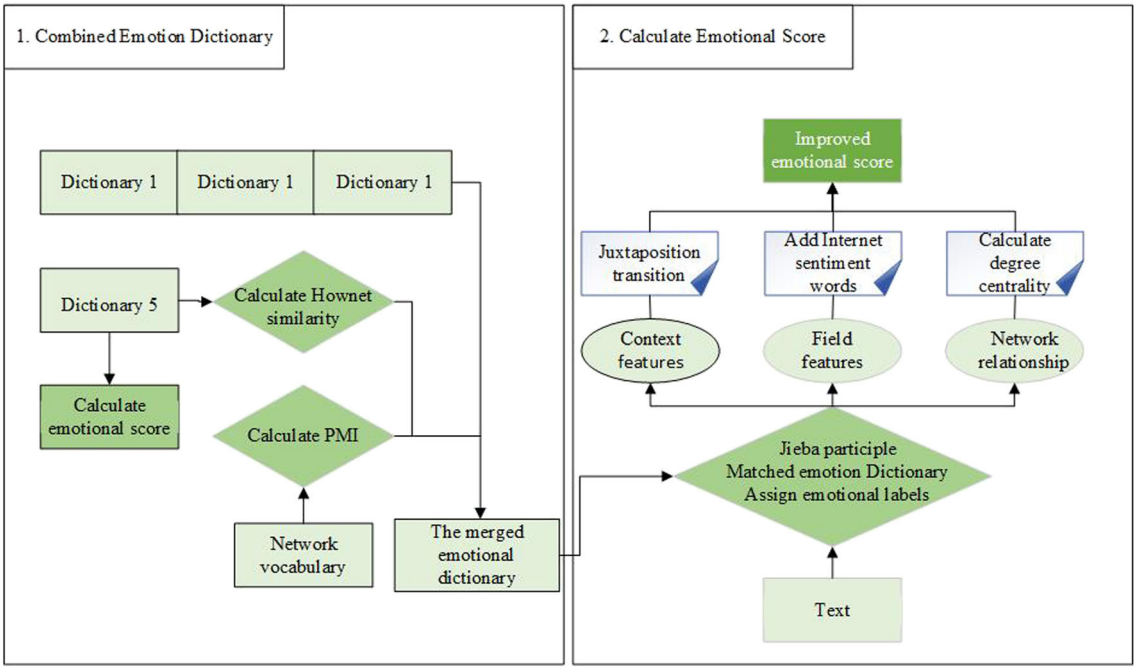

FIGURE 3 Sentiment analysis framework

analysis methods can improve the accuracy of sentiment classification (Eliacik & Erdogan, 2018; Wu et al., 2016).

2. The process of CDCSA constructing sentiment features

Juxtaposition and transition sentiment words in Chinese expressions respectively have the same and opposite sentiment tendencies. These words maintain the same or opposite sentiment tendency due to juxtaposition and transition. Simultaneously, the emotion words in the emotion dictionary can be divided into the following two parts: domain-independent and domain-related emotion words. Compared with the domain-independent emotion words, the domain-related emotion words express clear emotion tendency.

The aforementioned reasons are considered, and a novel sentiment analysis method CDCSA is introduced. The novel method is divided into the following three steps. (1) The commentating and negative thesaurus of Tsinghua University and HowNet sentiment dictionary is incorporated with the similarity degree. The inclusion of network words and the identification of the sentiment tendency of network words is achieved by using the pointwise mutual information algorithm. (2) The extraction of emotion words, computation of degree centrality between emotion words based on social network analysis, and identification of juxtative and transitional relationships in a corpus as constraints are realized through syntactic analysis and emotion dictionary. (3) The network relationship between sentiment words and constraints is utilized as the basis for the final sentiment score computation. Figure 3 presents the framework of sentiment analysis. The contextual relationship between sentiment words and the network relationship between documents and sentiment words are utilized in this paper to establish a domain sentiment dictionary.

3.3.2 | TextRCNN extracts deep features

1. The principle of textRCNN

The TextRCNN model uses a bidirectional recurrent neural network to learn contextual semantic features, and uses a max pooling layer for feature selection to finally complete this classification task. The model combines the advantages of convolutional neural networks and time series neural networks. As shown in Figure 4, first, word embedding is performed on the microblog text; second, the word vector is encoded from both positive and negative directions with a bidirectional long-short-term memory network model (BiLSTM) to mine long-distance dependencies and contextual semantic knowledge (Huang et al., 2021; Yang et al., 2022).

Its calculation equation is as follows.

$$\overrightarrow{h_r}(\mathrm{x_i}) = f\left(w_\mathrm{r} * \overrightarrow{h_r}(x_{\mathrm{i}-1}) + w_\mathrm{sr} * e(x_{\mathrm{i}-1})\right) \tag{2}$$

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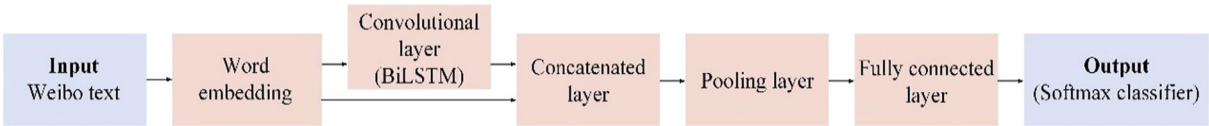

FIGURE 4 The neural network structure of TextRCNN model

$$\overleftarrow{h_l}(x_i) = f\left(w_l * \overleftarrow{h_l}(x_{i+1}) + w_{sl} * e(x_{i+1})\right) \tag{3}$$

$$t_i = \tanh\left(w_1 * \overrightarrow{h_r}(x_i) + w_2 * \overleftarrow{h_l}(x_i)\right) \tag{4}$$

In the formula, $\overrightarrow{h_r}$ represents the semantic vector of the LSTM layer to the right of the ith word, $\overleftarrow{h_l}$ represents the LSTM layer semantic vector of the ith word to the left, corresponding to the contextual feature information of the text; $e(x_i)$ represents the word vector of the ith word; w_r, w_l, w_{sr}, and w_{sl} represent the corresponding weight parameters; t_i is the final output of the bidirectional LSTM layer. It is input to the max pooling layer for feature selection, and the final classification result is obtained through the softmax classifier of the fully connected layer.

2. The process of textRCNN learning deep features

The Weibo posts not only indicate public sentiment and opinions but also demonstrate knowledge regarding the fluctuating trend of soybean futures prices. We use TextRCNN to extract deep features about price fluctuations from Weibo texts, including text labeling, text representation and feature extraction.

First, label the text by comparing the relationship between the current price and the price of the previous day, as shown in Equation (5), $price_{today}$ is the price of the current day, and $price_{yes}$ is the price of the previous day.

$$Labeltrend = \begin{cases} 1, price_{today} - price_{today} > 0 \\ -1, price_{today} - price_{today} < 0 \end{cases} \tag{5}$$

Second, use Word2vec to perform Word embedding on the text to obtain the semantics, grammar and structure information of the text.

Finally, TextRCNN trains the model, obtains the relationship between text features and target classification, and automatically learns rich and effective deep features.

3.4 | Factor selection based on multiple stepwise regression analysis

Stepwise regression is the introduction of all independent variables into the regression equation individually based on their role, importance, or contribution to the dependent variable Y. F test and t test should be respectively conducted for each introduction of an explanatory variable and each selected explanatory variable. An originally introduced explanatory variable is eliminated when it loses its importance due to the introduction of a later explanatory variable. The regression equation should only include significant variables before the introduction of each new variable. The aforementioned process is repeated until no significant explanatory variables are chosen or removed from the regression equation to create an optimal regression equation for predicting or controlling the dependent variable. Text features of social media and macroeconomic variables are the chosen regression parameters in this paper. The effect of measurement unit selection of indicators on regression results must be removed. Thus, the final multiple regression model is acquired by normalizing all indicators in the formula as presented in Equation (6).

$$\begin{aligned} Y = {} & a_1X_1 + a_2X_2 + a_3X_3 + a_4X_4 + a_5X_5 + a_6X_6 + a_7X_7 \\ & + a_8X_8 + a_9X_9 + a_{10}X_{10} + a_{11}X_{11} + a_{12}X_{12} + a_{13}X_{13} \\ & + e \end{aligned} \tag{6}$$

where Y denotes soybean futures price; X_1 denotes open interest; X_2 denotes volume; X_3 denotes the spot price of soybeans; X_4 denotes the US soybean futures price; X_5 denotes the exchange rate; X_6 denotes the spot price of WTI crude oil; X_7 denotes the closing price of yellow corn futures; X_8 denotes yellow corn holdings; X_9 denotes edible oil spot index; X_{10} denotes WTI crude oil futures price; X_{11} denotes sentimenttrend1; X_{12} denotes imsentimenttrend1; X_{13} denotes label trends; a_1–a_{13} denote the corresponding coefficients of each variable; e is constant.

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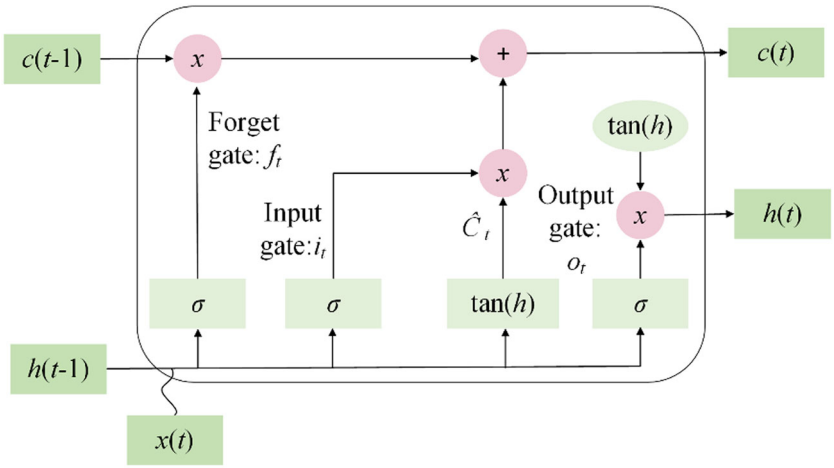

FIGURE 5 LSTM algorithm principle

3.5 | Long short-term memory (LSTM) algorithm principle and forecasting design

3.5.1 | LSTM algorithm principle

LSTM, a specific form of recurrent neural network (RNN), comprises memory modules. The long-term dependence problem of RNN is addressed by LSTM by raising the threshold RNN. LSTM also solves the long-term memory deficiency, gradient disappearance, or gradient expansion caused by numerous multiplications of the Jacobian matrix (Chuluunsaikhan et al., 2020). Therefore, long-distance timing information can be effectively used by the RNN. Figure 5 shows the specific structure. Each control door is defined as follows:

$$i_t^j = \sigma(\mathrm{U}_i x_t + w_i h_{t-1} + b_i)^j \tag{7}$$

$$f_t^j = \sigma(U_o x_t + w_o h_{t-1} + b_o)^j \tag{8}$$

$$c_t^j = f_t^j c_{t-1}^j + f_t^j \tilde{c}_t^j \tag{9}$$

$$\tilde{c}_t^j = tanh(U_c x_t + w_c h_{t-1} + b_c)^j \tag{10}$$

$$h_t^j = o_t \tanh\left(c_t^j\right) \tag{11}$$

3.5.2 | Design of a hybrid forecasting model based on LSTM (TSH-LSTM)

The design process of the proposed TSH-LSTM prediction model is as follows.

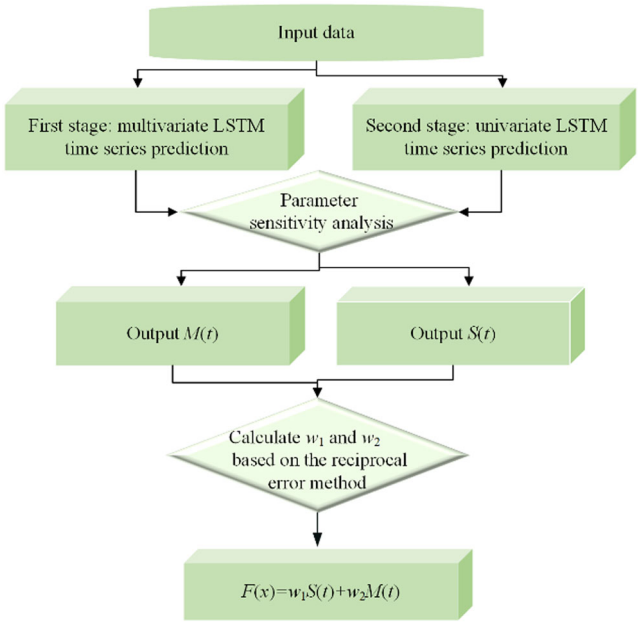

FIGURE 6 The framework of the TSH-LSTM

First, a univariate time series prediction of soybean futures price is presented by using LSTM.

Second, the development of a multivariable time series prediction model based on the regression results in Section 3.2 is realized through LSTM. Third, weight assignment to the two aforementioned prediction results through the reciprocal error method facilitates the acquisition of the final forecast value of soybean futures price. The predicted value of the univariate model at time t is assumed to be S(t), while that of the multivariate model is $M(t)$, $t = 1, 2, ..., n$. The two single models are then combined through the reciprocal error method. The integrated hybrid prediction model is defined as follows:

$$\mathrm{F}(t) = w_1 S(t) + w_2 M(t) \tag{12}$$

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$$w_1 = \frac{\varepsilon_M}{\varepsilon_S + \varepsilon_M} \tag{13}$$

$$w_2 = \frac{\varepsilon_S}{\varepsilon_S + \varepsilon_M} \tag{14}$$

where ε_S and ε_M, respectively, denote the average relative errors of univariate and multivariate models, and $F(t)$ indicates the hybrid predicted value at time t. The weight is computed through the reciprocal error method presented in Equations (13) and (14). A reduction in the average relative error of the entire hybrid model is realized through the assignment of a large weight to the model with a small average relative error, and an accurate predictive value is acquired. Figure 6 demonstrates the procedures of the proposed TSH-LSTM.

The widely used indicators, that is, root mean square error (RMSE), mean absolute percentage error (MAPE), and symmetric mean absolute percentage error (SMAPE), are used to assess the prediction performance of the model effectively (Wu et al., 2021). The equations are as follows:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{15}$$

$$\mathrm{MAPE} = \frac{1}{\mathrm{n}}\sum_{\mathrm{i}=1}^{\mathrm{n}}\left|\frac{\mathrm{y_i} - \hat{\mathrm{y}}_\mathrm{i}}{\mathrm{y_i}}\right| \tag{16}$$

$$\mathrm{SMAPE} = \frac{100\%}{n}\sum_{i=1}^{n}\frac{|\hat{y}_i - y_i|}{(|\hat{y}_i| + |y_i|)/2} \tag{17}$$

4 | EMPIRICAL EVALUATION

Experiments and evaluations are performed on the model in four steps to validate the proposed TSH-LSTM of the soybean futures price. The first step involves data collection and preprocessing, which includes the selection and description of numerical and text data. The second step is the creation of sentiment and deep features on Weibo text. Stepwise regression analysis is utilized in the third step to identify the substantial influence of social media features on soybean futures prices, and the analysis results are used in the selection of the prediction model input. The establishment of TSH-LSTM of soybean futures price based on LSTM occurs in the fourth step. The subsequent sections provide discussions and evaluations of the results of each phase.

4.1 | Data collection and description

Among the prices of major agricultural products, soybean price in China has the highest degree of marketization and has a high dependence on foreign countries. Thus, additional attention to price fluctuations is necessary. Therefore, two types of data, namely, structured numerical data and text data on social media, are chosen in this paper based on the existing studies on agricultural product futures price prediction theory and empirical literature (Table 1). Ten economic variables are also selected from the following three aspects: soybean fundamentals, futures market, and economic environment. Data from the National Bureau of Statistics, National Agricultural Product Cost and Income Data Collection, and the World Bank are used as bases for the prediction of soybean futures prices in China. Blog posts on Weibo regarding soybean futures and the soybean futures-related information on China Food are samples of text data on social media. Data collection from July 29, 2011, to July 19, 2021 was conducted using a web crawler based on Python 3.7. The current study relies on daily data analysis, except all closing prices with a daily volume of 0 and limited text data, which finally maintains 795 valid data. Model training uses the first 85% of the original data set as the training set and the remaining 15% as the test set to evaluate model performance. Meanwhile, the network structure is identified by dividing 15% of the data from the training set as the verification set. Effective model convergence to obtain the true value from the prediction can be realized by normalizing each training and test data and employing the formula to facilitate denormalization at the end of the prediction process (Table 2).

4.2 | The results of text feature mining

4.2.1 | Results of CDCSA constructing sentiment analysis

A sentiment analysis method combining sentiment dictionary and degree centrality is introduced in this paper considering the contextual and sentiment intensity features of sentiment words, and the collected text data are used for verification. A comparison of the performance of the proposed sentiment analysis method with that based on the sentiment dictionary is executed through the aforementioned method. The analysis results based on the emotion dictionary are presented in Table 3. The CNKI Emotion Dictionary is used to obtain emotion words from the provided text. A Weibo is positive when it comprises more positive emotion words than negative emotion words and vice versa. The degree of centrality of

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TABLE 1 Data type and variable introduction

Data type	Data source	Variable name	Selection basis
Numerical data	Soy fundamentals	Edible oil spot index, Yellow corn futures closing price, Yellow corn holdings	Hernandez et al., 2014; Marfatia et al., 2022.
	Economic environment	Soybean spot price, WTI crude oil spot price, WTI crude oil futures prices, Exchange rate	Xiong & Bao, 2020.
	Futures market	Open interest, Volume, U.S. soybean futures prices	Ahumada & Cornejo, 2016b.
Text data	Social media	Sentimenttrend, Imsentimenttrend, Labeltrend	Behavioral Finance Theory (Dolan, 2002).

TABLE 2 Descriptive statistics of variables

	N	Min	Max	Mean	Standard deviation
Soybean futures closing price	794	3122	6293	4507.68	781.018
Open interest	794	71,939	381,830	174486.40	64502.372
Volume	794	6490	969,885	141138.63	102270.958
Soybean spot price	794	3349.47	5227.89	4097.4419	515.88089
U.S. soybean futures prices	794	17.75	38.98	25.0781	6.06839
Exchange rate	794	6.0503	7.1681	6.617362	0.3231890
WTI crude oil spot price	794	8.910	110.62	60.128	21.456
WTI crude oil futures price	794	11.57	109.77	60.0847	20.99642
Yellow corn futures closing price	794	1394	2881	2264.82	323.385
Yellow corn holdings	794	132,498	2,246,161	916651.56	531421.148
Edible Oil Spot Index	794	307.93	668.32	420.0428	82.81389
Sentimenttrend	794	−5	141	94.35	37.167
Imsentimenttrend	794	−5	138	83.10	30.383
Labeltrend	794	−5	78	31.25	14.572
Number of valid cases (in a row)	794				

each sentiment word is then computed by the social network analysis method as presented in Table 4. The importance of the sentiment word is high when the degree center is large. The network relationship between the sentiment words is demonstrated in Figure 7.

The sentiment curve from the two sentiment analysis methods and the price trend curve of soybean futures are presented in Figure 8. Results indicate that the sentiment curve based on the CDCSA method demonstrates a high degree of fit with the price curve. Table 5 reveals that the Accuracy, Precision, Recall, and *F* values of the CDCSA method are 0.8, 0.83, 0.81, and 0.82, respectively. Thus, the classification performance of CDCSA is superior to the method based on the sentiment dictionary.

4.2.2 | Results of TextRNN extracting deep features

In this paper, TextRCNN is used to extract the depth features of the microblog text. Transformer, TextRNN_Att, TextCNN, DPCNN, and TextRNN are used as comparison models, which are extensively utilized in the field of natural language processing (Huang et al., 2021; Yang et al., 2022. Anaconda + Python3.7 + Pytorch1.13.1 is

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TABLE 3 Results of sentiment analysis

Positive words	Negative words	Positive score	Negative score	The total score
	Short, serious	0	−2	−2
Important, growing, positive, steady	Not as good as Serious	1	0	1
The main force, strong, dazzling	Short	3	−1	2
Talent, top notch	Inexplicable, squeezed, greedy, expansive	0	−2	−2
Benefit, increase	Rupture	2	−1	1
Beyond	The hype, the tension, the squeeze	0	−2	−2

TABLE 4 Co-occurrence relationship and degree centrality of emotional words

	Degree of centricity	Rose	Fall	Rebound	Shock	Sharp	Drop	Back	Strong
Rose	107	227	32	43	30	16	18	16	24
Fall	95	32	145	28	24	16	30	14	12
Rebound	92	43	28	156	35	15	16	14	21
Shock	81	30	24	35	221	17	18	24	21
Sharp	70	16	16	15	17	84	7	9	7
Drop	67	18	30	16	18	7	90	11	15
Back	72	16	14	14	24	9	11	85	5
Strong	67	24	12	21	21	7	15	5	100

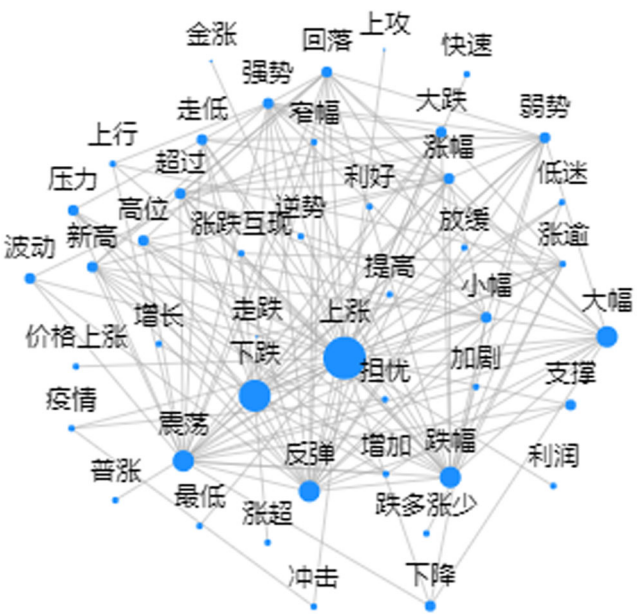

FIGURE 7 Social network of emotional words

the experimental environment, and PyCharm is the development tool. Table 6 presents the model and parameter settings in this paper. The current study aims to find a suitable classification model suitable and deliver clarifications for other text classification tasks.

The accuracy, recall, and F values of all classifiers are presented in Table 7. Results reveal that textRCNN had the highest accuracy and recall after 10-fold cross-validation, thus indicating its effectiveness in price fluctuation feature analysis in textual data. The advantages of RNN and CNN models contribute to the effectiveness of TextRCNN, thus allowing not only learning of local text features but also the sequence-dependent representation of text. TextRNN_Att and Transformer models have obtained optimal results in numerous natural language processing tasks. However, these models are relatively complex and need a considerable amount of data for model training to realize the expected performance. Moreover, sufficient information regarding the factors affecting prices may not be obtained through social media messages. Popular events in the market frequently gain the attention of social media platforms, and possible fluctuations in soybean futures prices can be attributed to variations in supply and demand or financial markets. The aforementioned factors that contribute to price variations may not be considered by social media. Therefore, information for predicting soybean futures prices is limited when only text data are used, and the integration of other financial market data sources into the futures price forecasting model is necessary to realize effective

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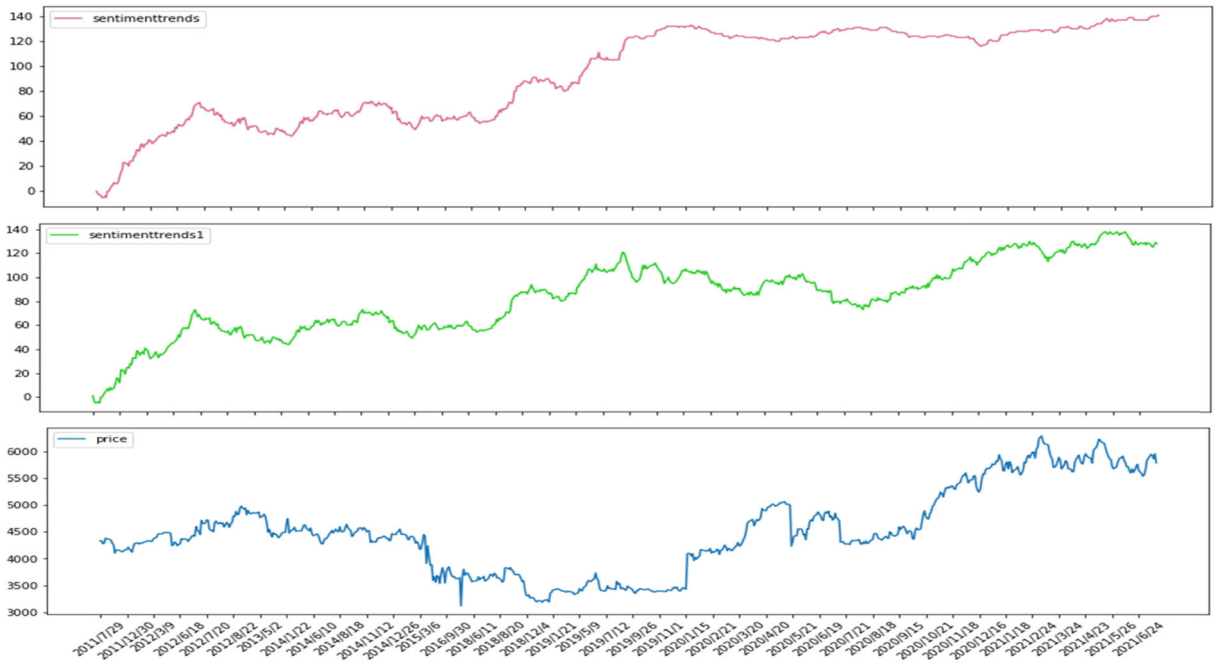

FIGURE 8 Sentiment curve and price trend

TABLE 5 Performance evaluation of sentiment analysis

Methods	Accuracy	Recall	*F* value	Precision
Sentiment dictionary	0.68	0.72	0.79	0.69
CDCSA method	0.8	0.83	0.81	0.82

TABLE 6 Deep learning model and parameter setting

Model	Parameter setting
Transformer	Dropout, 0.01; num_classes, 2; num_epochs, 100; batch_size, 64; pad_size, 32; learning_rate, 0.01; dim_model = 300; hidden = 1024; last_hidden = 512; num_head = 5; num_encoder = 2.
TextRNN_Att	Dropout, 0.01; num_classes, 2; num_epochs, 100; batch_size, 64; pad_size, 32; learning_rate, 1e−3; hidden_size, 64; num_layers, 2; hidden_size2, 32.
TextCNN	Dropout, 0.01; num_classes, 2; num_epochs, 100; batch_size, 64; pad_size, 32; learning_rate, 0.01; filter_sizes, 3; num_filters,16.
DPCNN	Dropout, 0.01; num_classes, 2; num_epochs, 60; batch_size, 64; pad_size, 32; learning_rate, 0.01; num_filters,16;
TextRNN	Dropout, 0.01; num_classes, 2; num_epochs, 20; batch_size, 64; pad_size, 32; learning_rate, 0.01; num_filters, 16; hidden_size, 128; num_layers, 2
TextRCNN	Dropout, 0.01; num_classes, 2; num_epochs, 100; batch_size, 64; pad_size, 32; learning_rate, 1e−3; hidden_size, 256; num_layers, 1

TABLE 7 Classification performance evaluation

	Acc	Precision	Recall	F1-score
Transformer	57.81%	0.5781	1.0000	0.7327
TextRNN_Att	53.12%	0.7059	0.3243	0.4444
CNN	46.32%	1.0000	0.1053	0.1905
DPCNN	60.00%	0.6154	0.9600	0.7500
TextRNN	57.81%	0.5781	1.0000	0.7327
TextRCNN	62.50%	0.6757	0.6757	0.6757

forecasting performance. The deep feature curve and the price trend of soybean futures are presented in Figure 9.

4.3 | Stepwise regression analysis results

The 13 factors presented in Section 4.1 are further screened through the stepwise regression method. The dependent and independent variables are soybean futures

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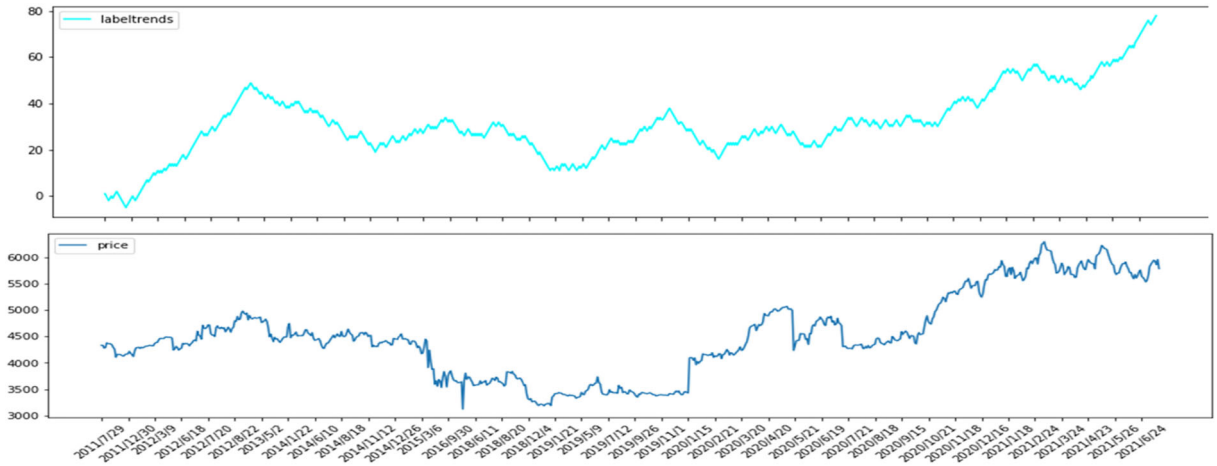

FIGURE 9 Trends in deep learning text classification and price curve

price and 13 influencing factors, respectively. The stepwise regression analysis was conducted using SPSS software, and retained variables are those that passed the F and t tests; otherwise, they were eliminated. Stepwise regression analysis finally identified the factors that substantially affect soybean futures price. The significance analysis results of the last six regression models are presented in Table 8. Model y(12) was used to identify the final factors. The model contains all factors, except the edible oil price, and the adjusted R^2 value is 0.922.

4.4 | Soybean futures price forecasting results

This stage involves the input process of text features (sentimental features, deep features) and financial market indicators to the soybean futures price prediction model. The proposed TSH-LSTM is compared to the univariate LSTM, multivariable LSTM, and XGBoost. The reason for choosing XGBoost as the benchmark model is that XGBoost (extreme gradient boosting) is a multithreaded implementation of gradient boosting decision tree (GBDT), which has the advantages of fast learning speed and high prediction accuracy (Chen & Guestrin, 2016). The existing research has demonstrated the advantages of XGBoost in agricultural product forecasting (Kang et al., 2020; Obsie et al., 2020). Therefore, XGBoost is chosen as the benchmark prediction model. The optimal parameters of each model are identified using the grid search method as presented in Table 9.

The time span of information acquisition is identified through the length of data segments in the training pool, which has a considerable impact on accuracy. A substantially small length will induce long-term time dependence loss, whereas a considerably large length will introduce unnecessary redundancy and computation. The parameter sensitivity of data segment length in the training pool of univariate and multivariate models is examined. Figure 10a and Figure 10b, respectively, demonstrate the price prediction values of the univariate and multivariate LSTM models with different Seq_length. The optimal performance of the two models is realized when the length is 7. This finding indicates the effectiveness of relevant information within 7 days for soybean futures price prediction. The results showed that, given that there are 7 days a week, investors generally pay more attention to information about the past week. Figure 11 demonstrates the errors of univariate and multivariate prediction models under different Seq _lengths, thus facilitating the evaluation of the influence of different Seq _lengths on the prediction results. The values of evaluation criteria under different Seq_length for the multivariate prediction and univariate models are respectively presented in Tables 10 and 11. The values of MSE, RMSE, MAE, and other indicators are the minimum when Seq_length is 7 for the two models.

Figure 12a presents the average relative error of the TSH-LSTM and every single model, whereas Figure 12b demonstrates the comparison between predicted and actual values. The proposed TSH-LSTM has higher prediction accuracy than the other three models and demonstrates a similar prediction curve to the actual curve. Satisfactory prediction accuracy is obtained by the univariate prediction model in the early step, but the prediction capability and error respectively decrease and increase with the increment of times. However, the multivariate prediction model showed a robust overall prediction performance and effective long-term prediction performance. Compared with the XGBoost model, the two prediction models based on LSTM are remarkably accurate. The proposed TSH-LSTM by the reciprocal

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TABLE 8 Significance analysis results

Variable	y(6)	y(7)	y(8)	y(9)	y(10)	y(11)	y(12)
X_{10}							−0.147** (0.057)
X_5						0.134*** (0.031)	0.151*** (0.031)
X_{12}					0.148*** (0.033)	0.167*** (0.033)	0.170*** (0.033)
X_{11}			−2.254(.010)	−0.046** (0.023)	−0.190*** (0.040)	−0.301*** (0.047)	−0.327*** (0.048)
X_1			−0.061*** (0.014)	−0.064*** (0.014)	−0.041** (0.015)	−0.028** (0.015)	−0.028* (0.015)
X_{13}		0.103*** (0.017)	0.080*** (0.018)	0.092*** (0.019)	0.061** (0.020)	0.107*** (0.022)	0.119*** (0.023)
X_2	0.065*** (0.013)	0.081*** (0.013)	0.101*** (0.013)	0.100*** (0.013)	0.105*** (0.013)	0.090*** (0.013)	0.087*** (0.013)
X_4	0.135*** (0.023)	0.125*** (0.023)	0.134*** (0.023)	0.131*** (0.023)	0.104*** (0.023)	0.106*** (0.023)	0.121*** (0.024)
X_8	−0.124*** (0.015)	−0.130*** (0.015)	−0.163*** (0.017)	−0.148*** (0.018)	−0.120*** (0.019)	−0.135*** (0.019)	−0.138*** (0.019)
X_6	−0.325*** (0.020)	−0.303*** (0.020)	−0.304(0.020)	−0.324*** (0.022)	−0.318*** (0.022)	−0.328*** (0.022)	−0.203*** (0.053)
X_7	0.371*** (0.021)	0.401*** (0.021)	0.408*** (0.021)	0.401*** (0.021)	0.379*** (0.021)	0.453*** (0.027)	0.460*** (0.027)
X_3	0.575*** (0.022)	0.471*** (0.028)	0.481*** (0.028)	0.495*** (0.028)	0.534*** (0.029)	0.525*** (0.029)	0.522*** (0.029)
Constant	−1.96 (0.011)	−2.45 (0.010)	−2.25 (0.010)	−2.13 (0.010)	−2.82 (0.010)	−1.62 (0.010)	−1.75*** (0.010)
N	795	795	795	795	795	795	795
Adj-R^2	0.912	0.916	0.918	0.918	0.920	0.922	0.922

Note: The values in brackets are T values, with *, **, and ** representing significance levels of 10%, 5%, and 1%, respectively.

TABLE 9 Prediction model parameters

Model	Parameter settings
XGBoost	n_estimators, 740; Gamma, 0; Subsample, 0.7; colsample_bytree, 0.7; learning_rate, 0.08.
Single variable LSTM	Epochs, 100; layers, 2; input_size, 1; hidden_size, 6; lr, 0.02; sequence_length, 7; batch_size, 32; dropout, 0.1.
Multivariate LSTM	Epochs, 100; layers, 2; input_size, 13; hidden_size, 32; lr, 0.01; sequence_length, 7; batch_size, 32; dropout, 0.1.

error method can incorporate the advantages of the two models and reduce the relative error considering the various advantages of univariate and multivariable models. Figure 12a shows that the average relative error of the proposed TSH-LSTM at each moment is small with minimal fluctuation, thus effectively reducing the prediction error and demonstrating an improved prediction effect. Compared with univariate prediction, multivariate prediction, and XGBoost models, the smallest MAE, MSE, and RMSE values, which are 0.068, 0.008, and 0.094, respectively, are observed in the proposed TSH-LSTM as presented in Table 12.

Moreover, the proposed TSH-LSTM demonstrated satisfactory adaptability to trend change points in the data, particularly strong robustness to numerous outliers, missing values, and trend changes. This model feature minimizes the adverse effect of data volatility on the forecast. The combination of univariate and multivariable prediction models with the error reciprocal method generates the mixed model. This model considers the advantages of two single models and compensates for the defect of large prediction errors of the single model. The enhanced prediction accuracy of soybean futures prices is attributed to the proposed TSH-LSTM.

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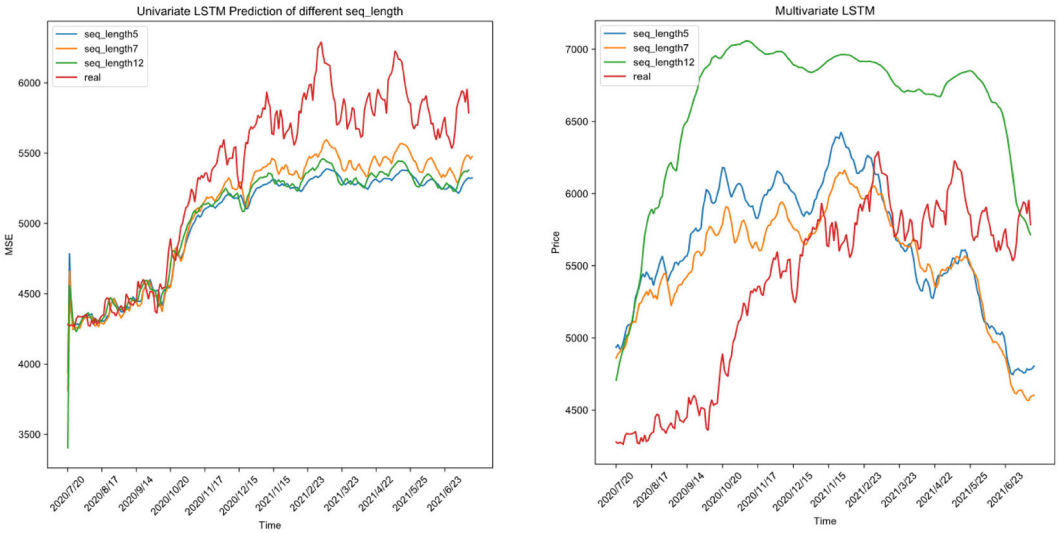
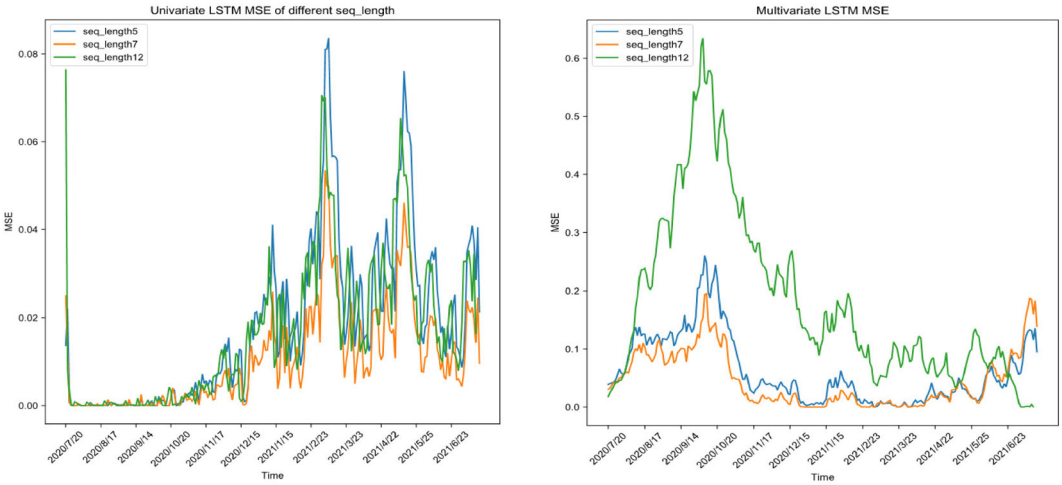


(a) Univariate LSTM Price Prediction Curve (b) Multivariate LSTM Price Prediction Curve

FIGURE 10 Price predictions for different Seq_lengths



(a) MSE of univariate LSTM (b) MSE of multivariate LSTM

FIGURE 11 Sensitivity analysis of prediction model

TABLE 10 Predictive performance evaluation of multi-LSTM model

	MSE	RMSE	MAE	MAPE	SMAPE	MRE
Seq_length_5	0.062	0.249	0.219	28.870	31.951	0.288
Seq_length_7	0.047	0.217	0.182	26.531	28.024	0.288
Seq_length_12	0.182	0.427	0.391	35.702	45.611	0.357

TABLE 11 Predictive performance evaluation of uni-LSTM model

	MSE	RMSE	MAE	MAPE	SMAPE	MRE
Seq_length_5	0.018	0.134	0.110	11.102	12.374	0.100
Seq_length_7	0.010	0.102	0.084	8.716	9.875	0.089
Seq_length_12	0.016	0.127	0.105	10.089	9.339	0.098

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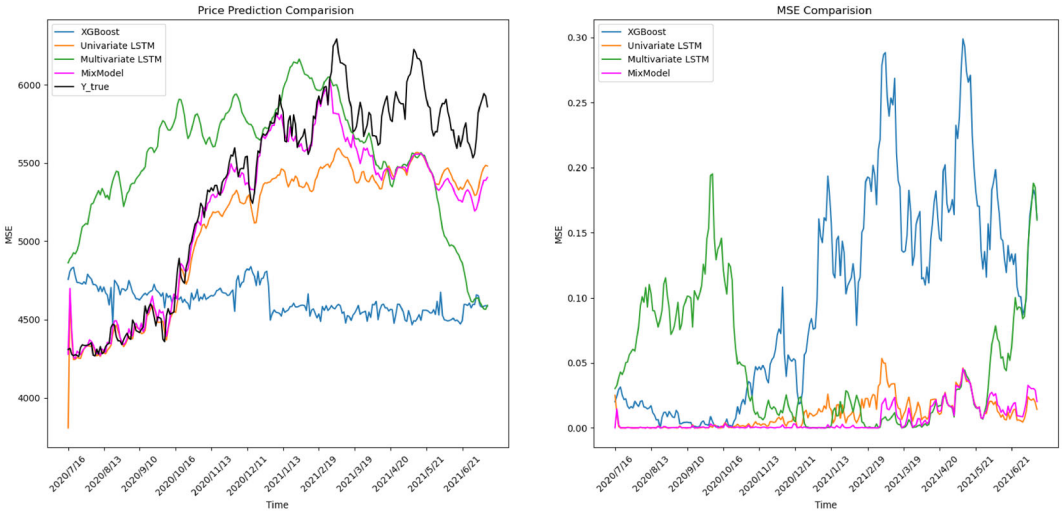

(a) Price forecast curve comparison (b) MSE comparision

FIGURE 12 Performance evaluation of TSH-LSTM

TABLE 12 Forecasting results of different models when Seq_length_7

	MSE	RMSE	MAE	MAPE	SMAPE	MRE
TSH-LSTM	0.008	0.094	0.068	5.891	6.141	0.058
Multi-LSTM	0.047	0.217	0.182	26.531	28.024	0.288
Uni-LSTM	0.010	0.102	0.084	8.716	9.875	0.089
XGBoost	0.098	0.314	0.276	59.862	44.314	0.59

5 | CONCLUSION AND FUTURE RESEARCH

An effective soybean futures price prediction model named TSH-LSTM is proposed based on the unique analysis of soybean futures and previous studies. Compared with previous studies, soybean futures price prediction is realized through the systematic use of multisource heterogeneous data and the time series model. Moreover, sentiment and fluctuation features are extracted from social media texts using advanced NLP technology, which extends the existing research on soybean futures price prediction.

The effects of social media feature on soybean futures prices are examined using the stepwise regression method. This method provides theoretical support and knowledge for the next step of employing the deep learning method for soybean futures price prediction.

5.1 | Theoretical contribution

a. An effective soybean futures price prediction model is constructed based on behavioral finance and information theories by considering massive data and the rapid development of information technology. The current study expands the theoretical system of existing soybean futures research. Previous studies mainly used time-series data and macroeconomic factors as bases for the prediction of soybean futures price, disregarding the effect of exogenous social media factors on the price. Social media is a platform for the public to express their feelings and opinions and is also regarded as a crucial part of knowledge sharing and distribution. Thus, particular problems can be addressed by extracting inherent knowledge in social media texts. Simultaneously, accurate prediction and theoretical support for soybean futures prices are provided by examining the emotional features and ups and downs features of prices (knowledge characteristics) on social media.
b. The combination of behavioral research and design scientific method improves the research system of soybean futures price prediction. Soybean futures price prediction using the deep learning method demonstrates substantially superior prediction performance to traditional models. However, the limited theoretical support of the model is attributed to its black-box

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mechanism. The effects of social media feature on soybean futures prices are examined through the combination of the econometric analysis and deep learning methods and the utilization of the stepwise regression method. This approach endows theoretical support for the prediction model based on deep learning.

c. This paper proposes an effective TSH-LSTM based on LSTM, which further improves the existing soybean futures price forecast model. The univariate time series prediction model has good short-term prediction performance but suffers from poor long-term prediction performance due to its limited capacity to learn features of a single sequence only and its weakness in handling changes. The multivariate prediction model is robust because it can learn the features of multiple factors. Therefore, a multivariable time series prediction model is established in two stages based on LSTM, and the prediction results of the two stages are combined to acquire the final mixed prediction model using the error reciprocal method.

5.2 | Practical contribution

a. A new method, which aims to establish social media text features of soybean futures, including emotional and price volatility features, is introduced in this paper. The accuracy of soybean futures price prediction is further enhanced through the mutual supplement of emotional and volatility features. Previous studies ignored knowledge features when analyzing social media text data. Compared with the benchmark model, the sentiment analysis method of integration degree centrality and the classification method of rising and fall based on textRCNN are both superior. A reference for social media text analysis is introduced by the proposed method.

b. This study provides an effective way of mining the values of digital and textual data. Thus, the proposed method can be applied to any field that uses the two data types. Analysis of other stocks or assets, usage of other information fusion methods, development of other trading rules, or utilization of other types of data from different sources are achieved in practice through the introduced framework.

5.3 | Limitations and future research

The introduced framework is only verified on soybean futures price data set due to space limitations. Thus, future research will mainly consider the following three aspects. (a) The use of a combination of the different characteristics of multisource data, such as daily trading data, unstructured data, and macroeconomic data on social media, and the time-frequency feature of decomposition forms a series of prices through the ITD method. (b) Extraction validation is conducted using Granger causality analysis. Assumptions regarding soybean futures prices in China can be obtained using social media features. The optimal feature subset is also identified to acquire new management enlightenment and theoretical support. (c) Examining the applicability of the proposed framework to other complex forecasting problems under uncertainty (Wang et al., 2019).

ACKNOWLEDGMENT

This work was partially supported by National Social Science Foundation of China (No: 20&ZD126).

DATA AVAILABILITY STATEMENT

The datasets used in the current study are available from the corresponding author on reasonable request.

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How to cite this article: An, W., Wang, L., & Zeng, Y.-R. (2022). Text-based soybean futures price forecasting: A two-stage deep learning approach. *Journal of Forecasting*, 1–19. https://doi.org/10.1002/for.2909

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